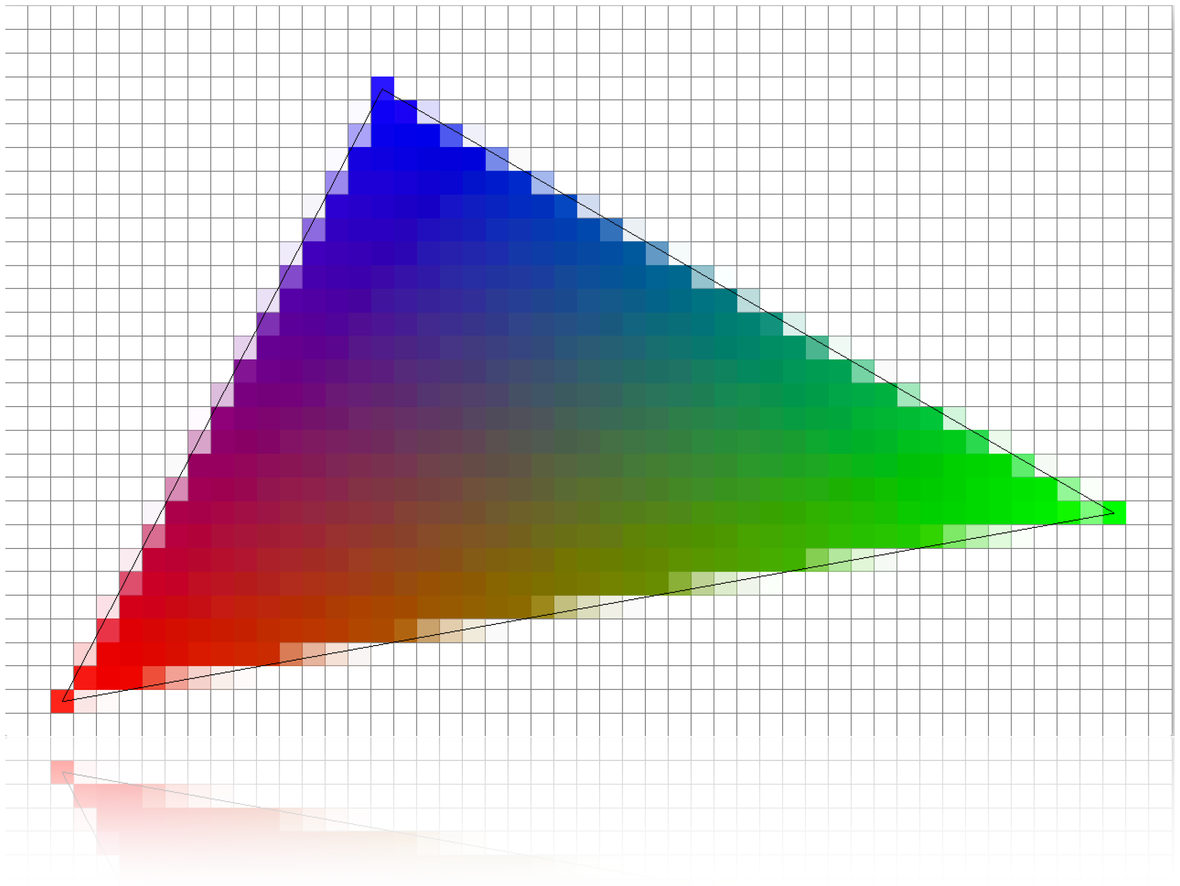
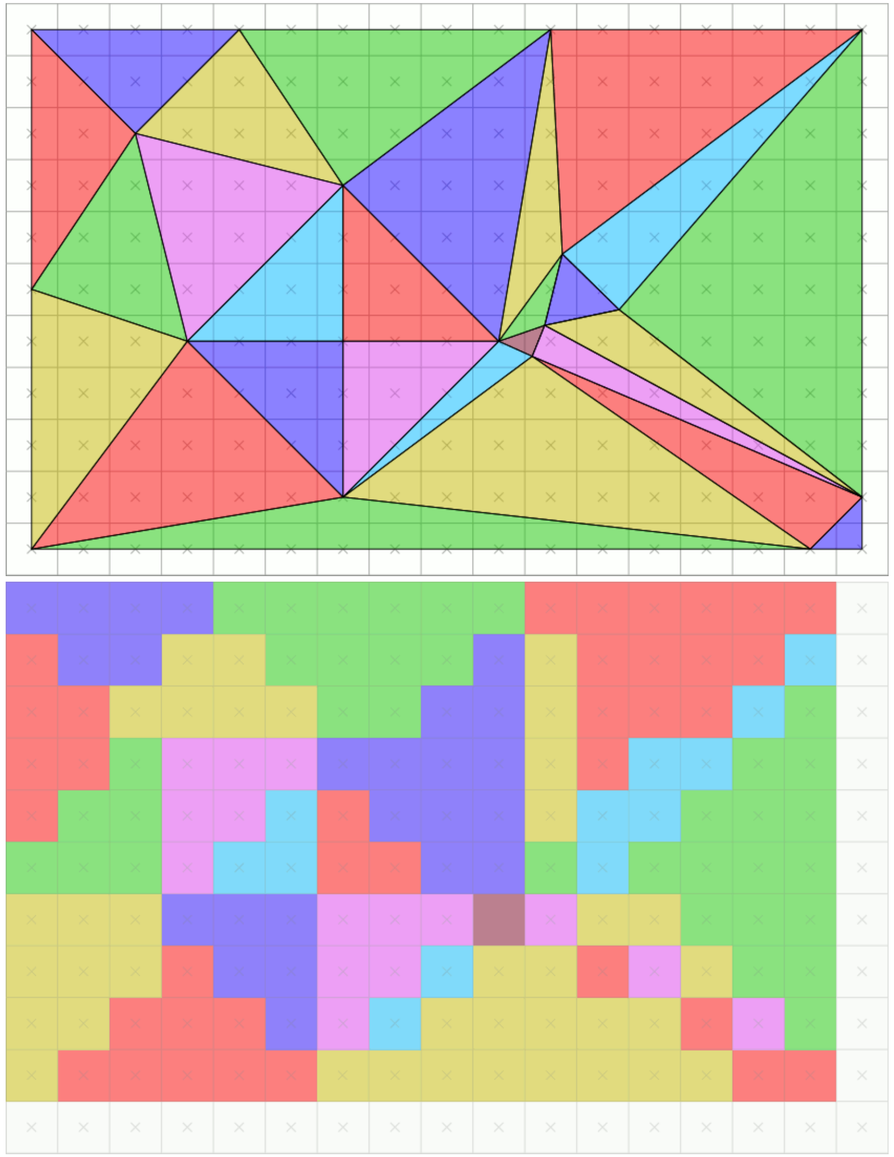
Rasterization

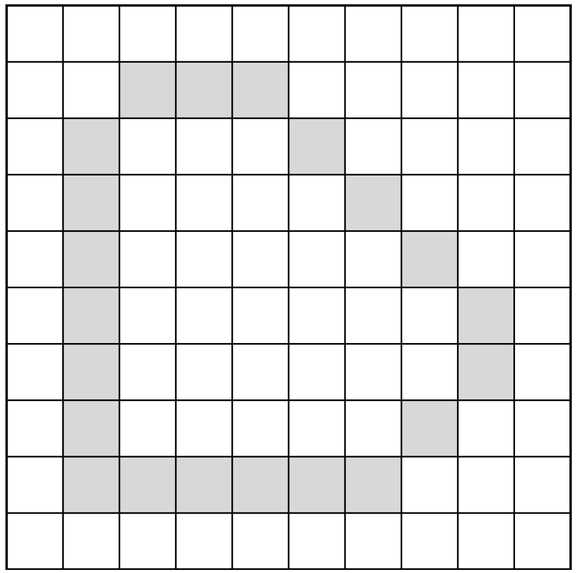


Fragments

- Rasterization is concerned with the formation of fragments of a raster image from vector graphics based geometric primitives
  - discretization of geometry after projection from 3D camera space to 2D screen space
  - a.k.a. **scan-conversion**
- Fragments are potential pixels
  - have a color attribute and location in screen space
  - carry depth information for visibility determination in 3D
    - multiple fragments can be generated for one pixel from all geometric primitives projecting onto the same pixel



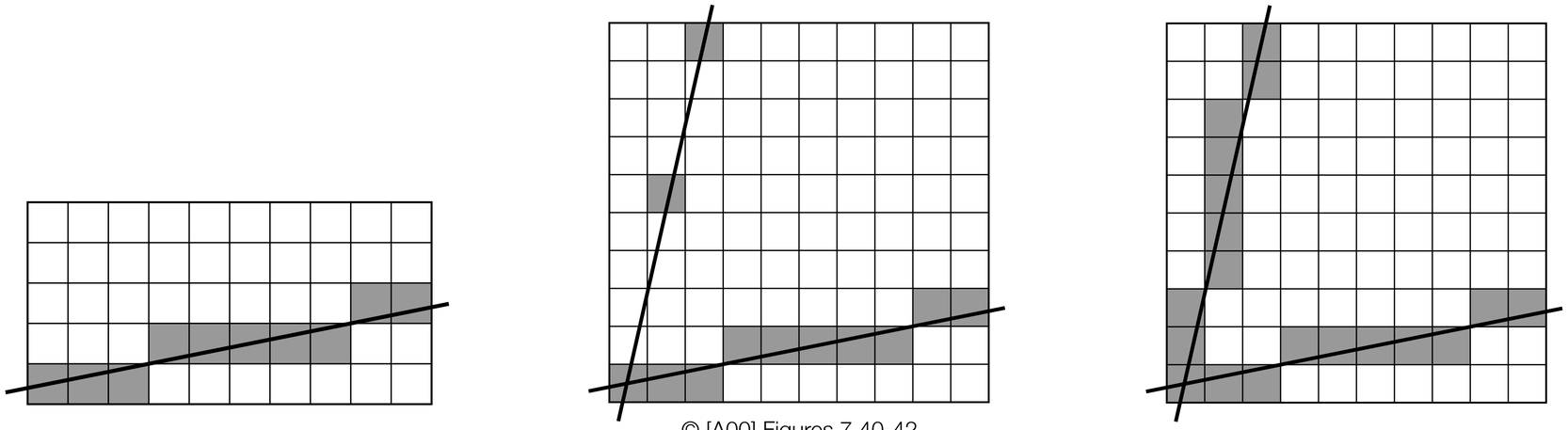
Scan-Conversion



© [A00] Figure 7.52

- Scan-conversion algorithms compute the coordinates of fragments that lie on or near a specified geometric primitive
  - must be **as fast as possible** as many million geometric primitives must be drawn in a small fraction of a second
- Drawing lines or filling polygons on a raster typically causes discretization artifacts, i.e. an aliasing problem
  - antialiasing techniques to remedy such problems

Line Symmetry



© [A00] Figures 7.40-42

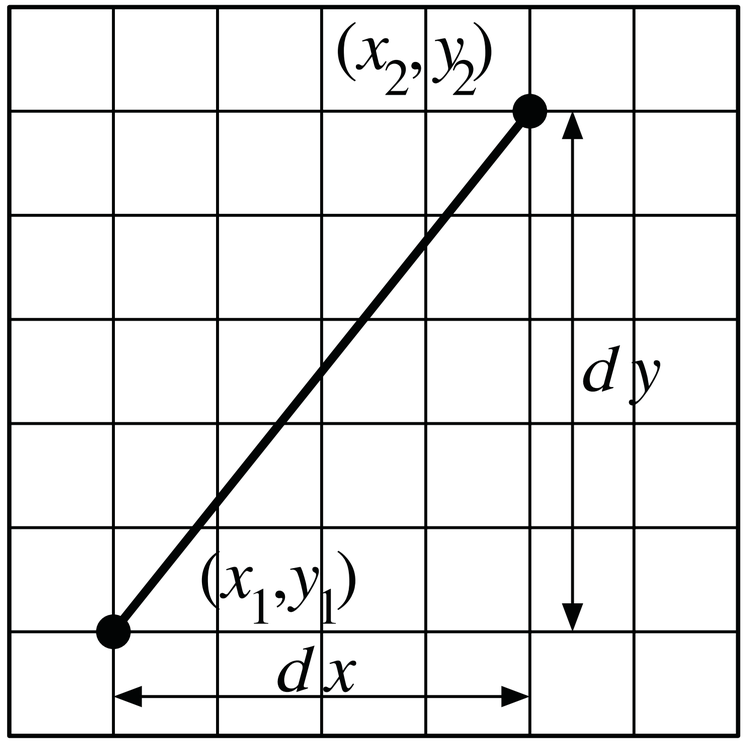
- For slopes m larger than 1.0, pixels in the y direction are skipped since we advance y by more than one pixel at a time
- Can easily be resolved by swapping the roles of x and y for larger slopes
  - adjust also for negative slopes
  - removes potential problems of horizontal or vertical lines $(m{=}\infty)$
- Implement algorithm for standard positive slopes m less than 1.0

Simple Line Drawing

- Assume slope $m = dy/dx$ in $[0,1]$ between two points
  - explicit line equation is $y = y_1 + m \cdot (x - x_1)$
- A unit change $dx = 1.0$ along x causes y to change by $dy = m$ between y_1 and y_2
- Fill pixels increasing in x from x_1 to x_2

```
float y = y1, m = (y2-y1)/(x2-x1);
for (x = x1; x <= x2; x++) {
  WritePixel(x, (int)floor(y+0.5), color);
  y += m;
}
```

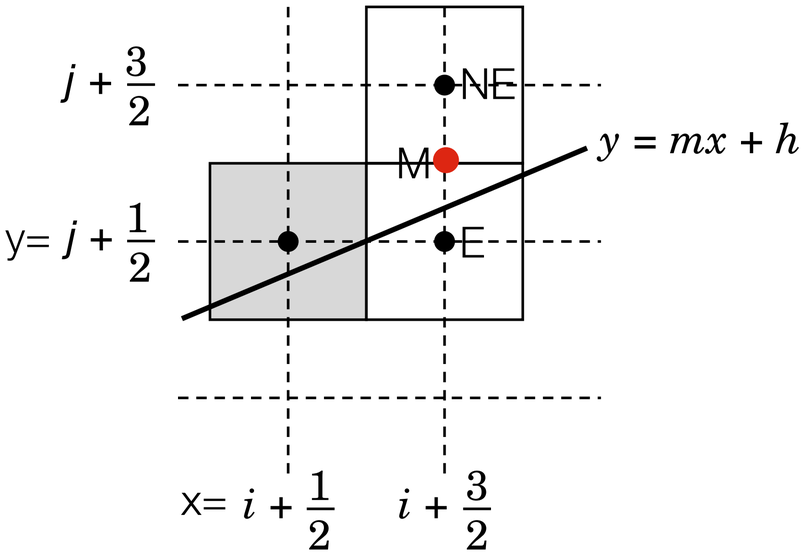
  - since float variables have limited precision, accumulation of m may cause cumulative error buildup for long lines



© [A00] Figure 7.39

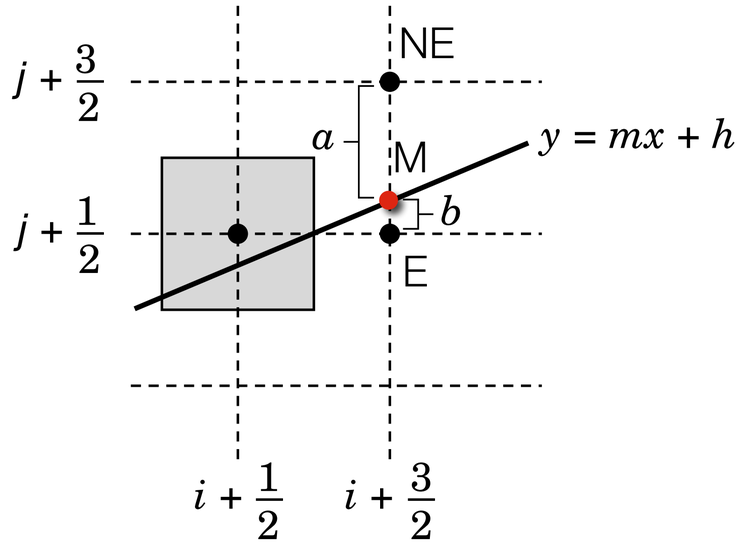
Bresenham's Algorithm

- Floating point and expensive rounding operations are time consuming if done in "gazillions"
  - Bresenham developed a scan-conversion avoiding floating point calculations
- Decide for each step if the next pixel:
  is to the East (E) or Northeast (NE)
  - decision is made based on the midpoint M between the $E = (x+1, y)$ and the $NE = (x+1, y+1)$ pixel
    - with the pixel centers $x = i+1/2$ and $y = j+1/2$
  - if M is above the given line, $M > y_1 + m \cdot ((x+1) - x_1)$ then the next pixel to fill is E, else it is the NE pixel
- All computations and decisions can be made using integers



© [A00] Figure 7.43

- Use of decision variable $d = a–b$ with $d > 0$ indicating E or $d < 0$ NE as the next pixel to be filled
  - going E: $a = a - 1 \cdot m = a - dy/dx, \quad b = b + 1 \cdot m = b + dy/dx$
    going NE: $a = a + 1 - m = a + 1 - dy/dx, \quad b = b - 1 + m = b - 1 + dy/dx$
  - but $\pm m = dy/dx$ is a float number, transform decision to only use integers
  - use non-rational decision variable $dx \cdot d = dx \cdot (a–b)$ without affecting the inequality $d > \text{or} < 0$
  - updating $dx \cdot d = dx \cdot a - dx \cdot b$, will use $\pm dy$ instead of $\pm\, dy/dx$
- We can update d incrementally either by $\Delta d_E = -2dy$ or $\Delta d_{NE} = 2(dx–dy)$, for going E or NE
  - going E: $dx \cdot d = dx \cdot a - dy - dx \cdot b - dy = dx \cdot a - dx \cdot b - 2dy = dx \cdot d - 2dy$
  - all integer only calculations to update
- Initialize d for the first step to $dx - 2dy$
  - for the first pixel at x_1 where the line intersects at y_1, set $a = 1$ and $b = 0$
  - first test at x_1+1 then uses $d = dx \cdot (a–b) + \Delta d_E = dx - 2dy$



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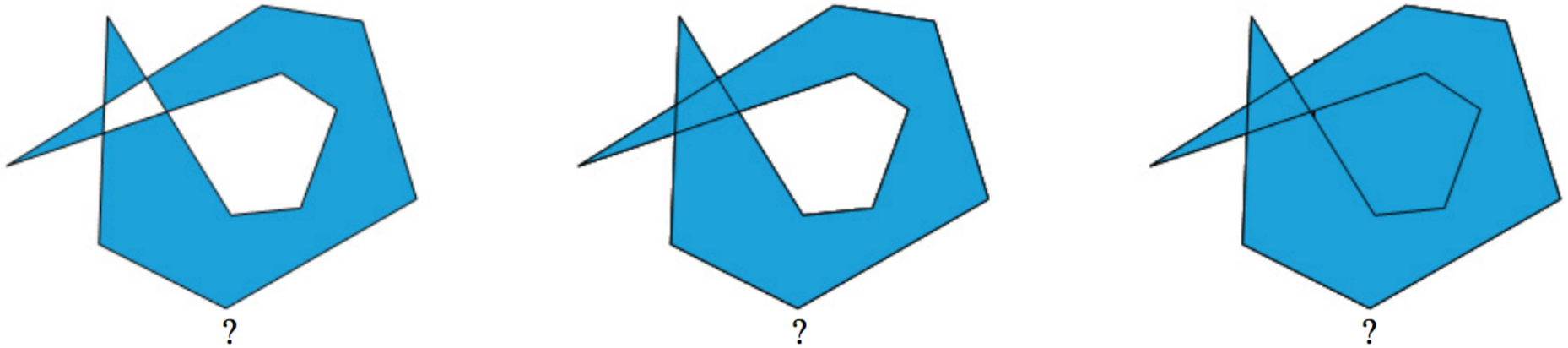
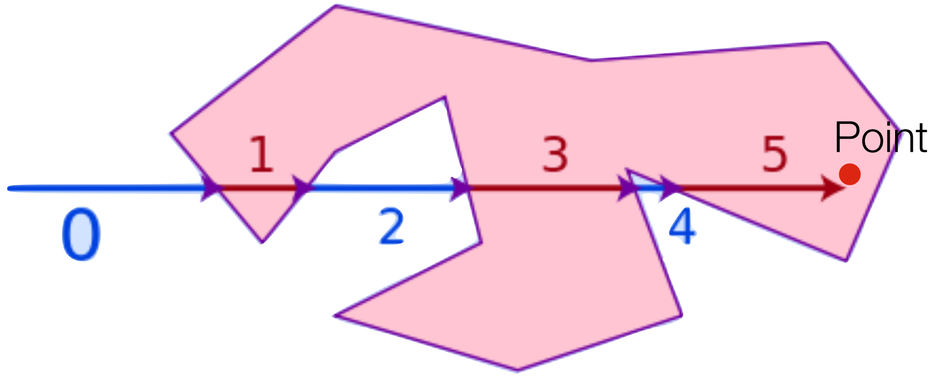
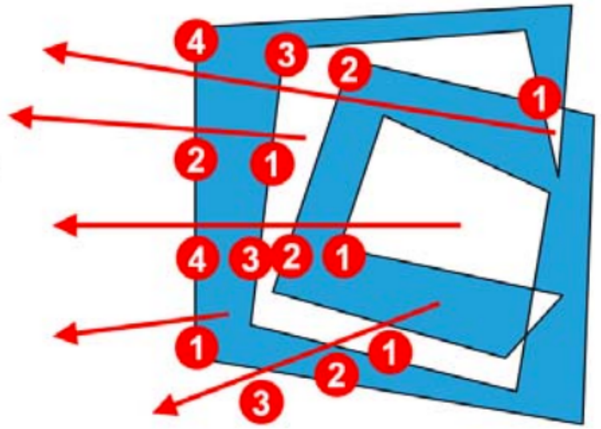
Polygon Filling



- *Main Question*: How to fill the interior of a polygon with a color
  - inside-outside are only well defined for flat simple polygons
- Typically: Fill interior of polygon with color during scan-conversion
  - convexity further simplifies scan-conversion filling process
  - non-convex and self-intersecting polygons complicate the process

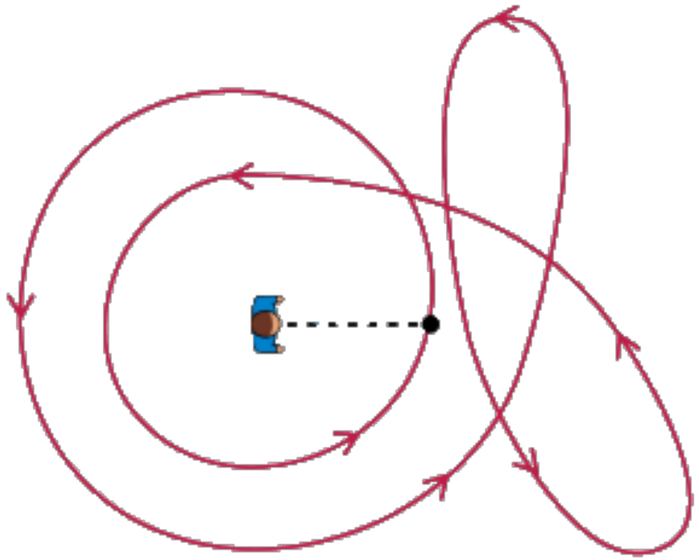
Odd/Even Inside/Outside Test

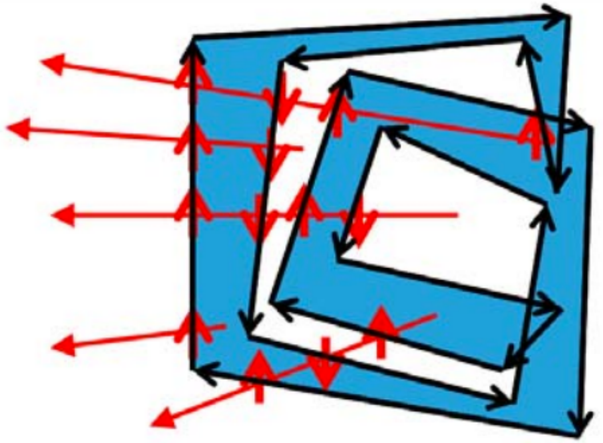




- Odd-even test is simple to implement
  - how many times a ray is crossing a polygon edge from infinity to a given query point
- Related to edge-intersection pairs in scan-line filling
  - test can be integrated with the rasterization of polygons

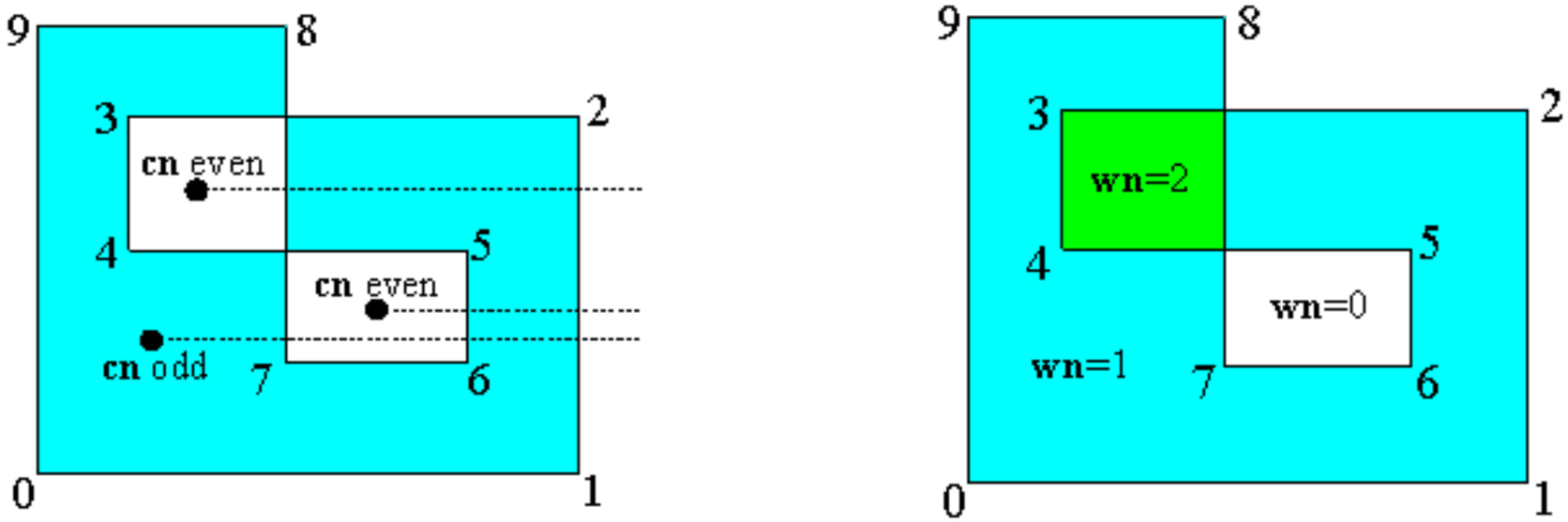
Winding Number Inside/Outside Test





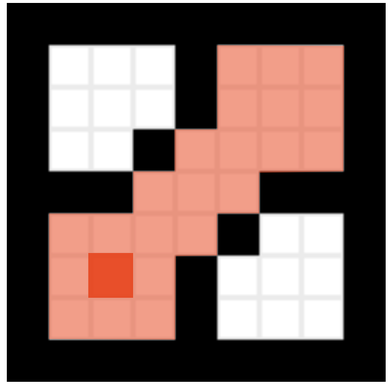
- Nonzero-Winding number rule: times circling around a point
  - point is outside if there is the same number of clockwise and counterclockwise directions
  - if winding number > 0 then point is inside
  - stable also for non-simple, non self-intersecting polygons

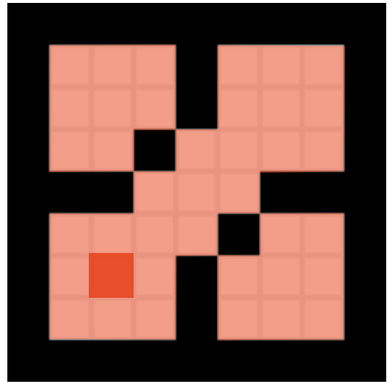
Odd/Even vs. Winding Number



Source http://geomalgorithms.com/a03-_inclusion.html

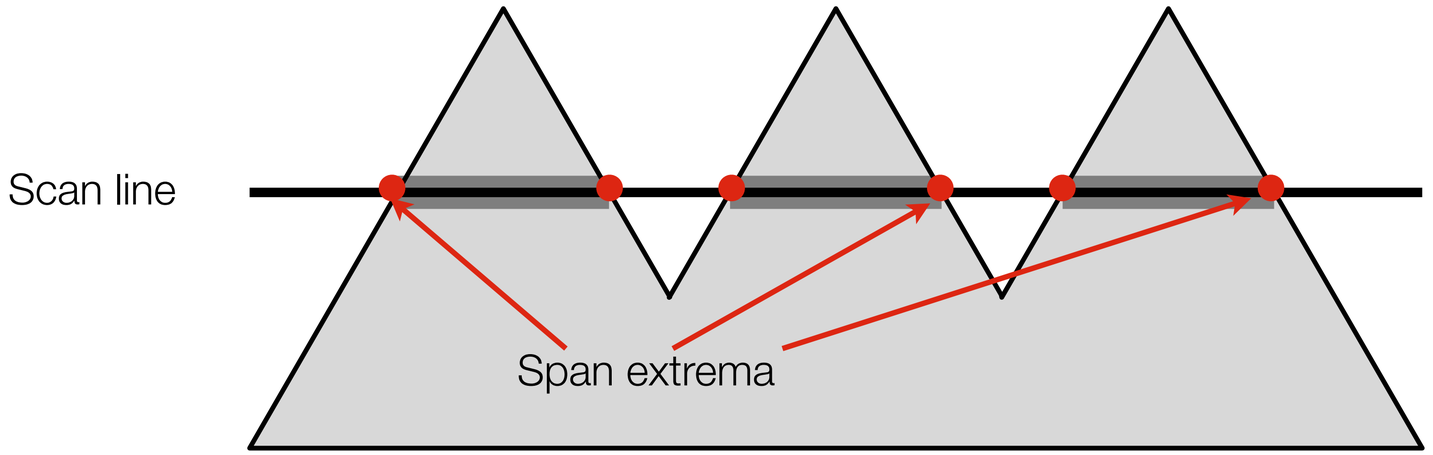
Pixel based Flood Filling





- Given an interior seed point, neighboring pixels can recursively be colored until reaching any prior rasterized polygon edge
  - recursion and duplicate work can be avoided using clever ordering and queuing strategies
- Different 4- and 8-way neighborhood filling results

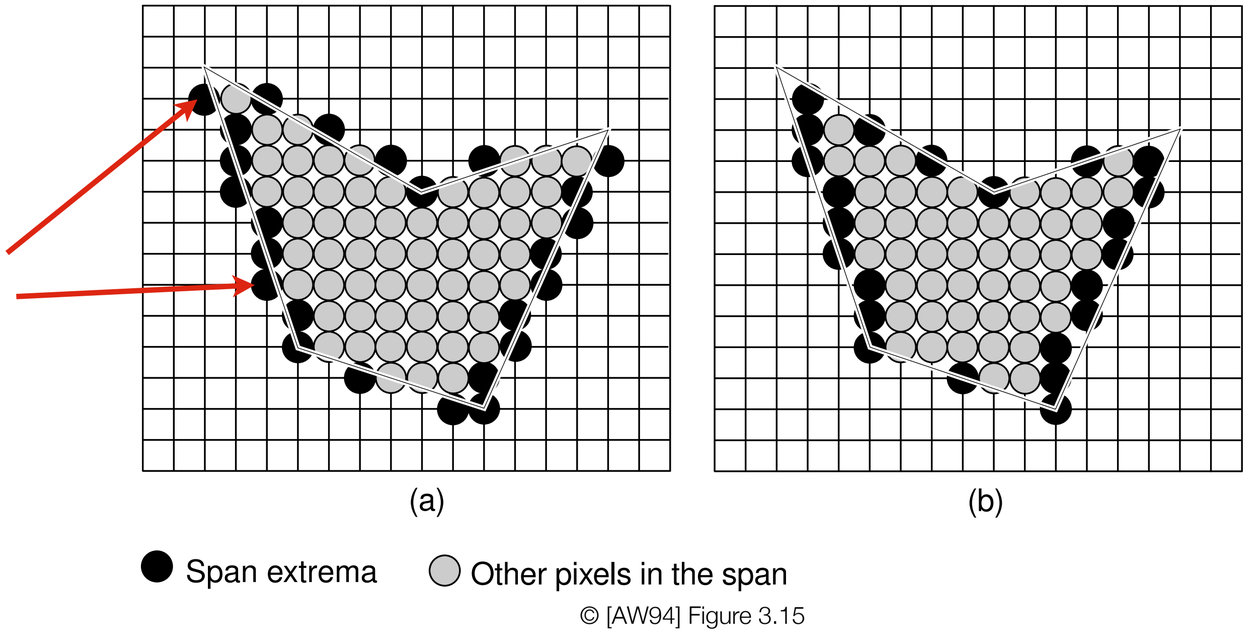
Scan-Line Filling



© [A00] Figure 7.53

- Fill **spans** of pixels on a scan line between the left and right edges of a polygon
  - works for convex as well as concave polygons
  - similar to odd-even test
- Span extrema required for each scan line during rasterization
  - maintained during polygon edge scan-conversion

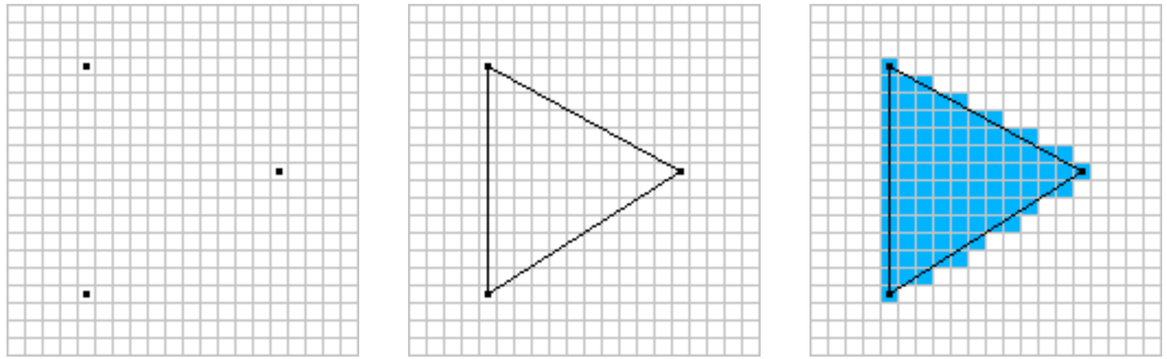
Polygon Boundaries



© [AW94] Figure 3.15

- Scan-conversion of polygon edges in general may produce pixels not necessarily inside the polygon
  - ambiguous coloring for adjacent polygons sharing the same edge
  - modified polygon edge scan-conversion algorithm required
    - pixels must be on the inside side of the polygon edge

Triangle Rasterization



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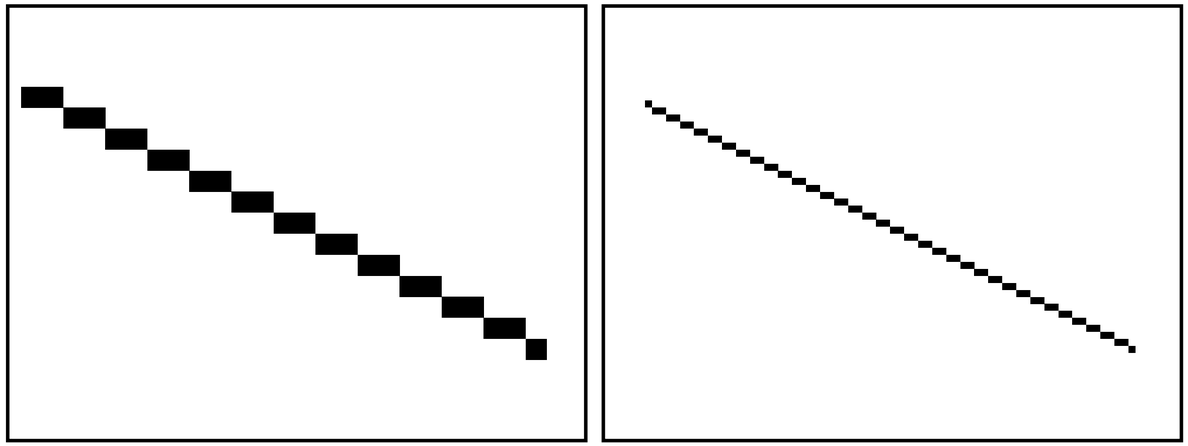
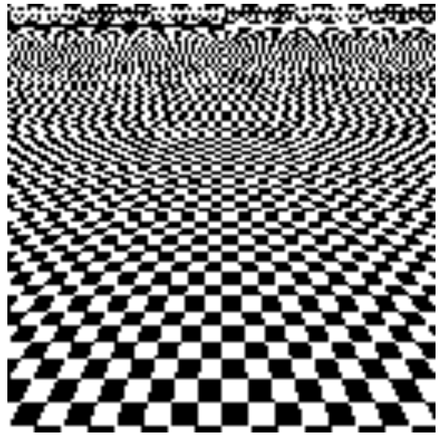
- Triangles are the preferred polygons to represent 3D shapes
  - guaranteed convex and planar polygons
- Rasterization of polygons can be achieved through rasterizing triangles
  - individual polygons broken up into triangles
- Exploit optimized triangle rasterization algorithm
  - determine the two edges of the triangle intersecting the scanline
  - loop through the y axis range of the triangle to calculate the horizontal spans
  - fill pixels between span endpoints
- Care has to be taken to not leave holes between adjacent triangles and to not rasterize any pixel more than once

Antialiasing



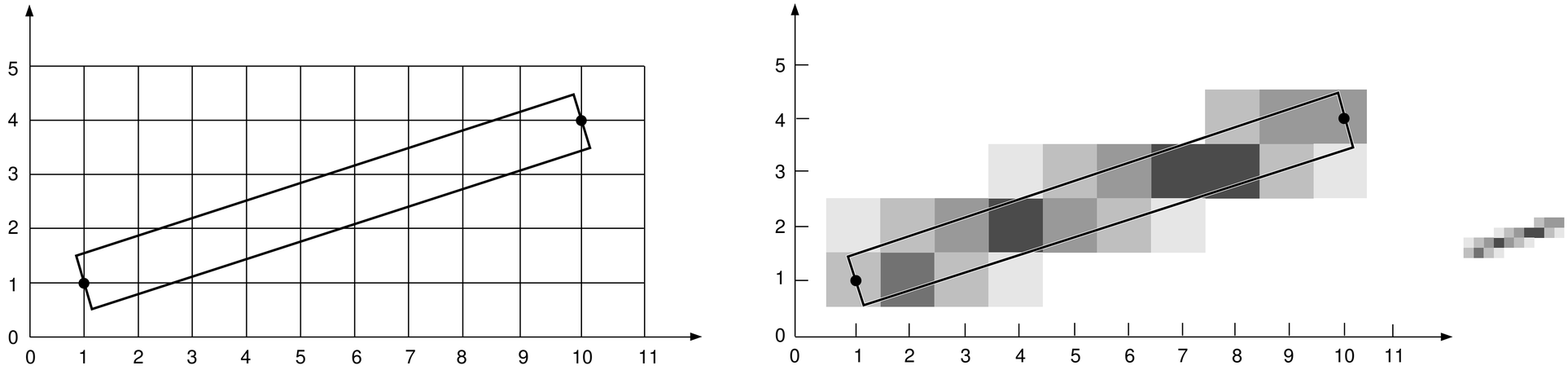
© [A00] Figure 7.59



- Aliasing problems are artifacts caused from discretization and sampling of a continuous signal
  - jagged staircase lines, moiré patterns
- Key is to remove signal components with higher frequency than sampling resolution
  - increasing the resolution does not eliminate the problem
  - variable pixel intensities can smooth out abrupt intensity changes

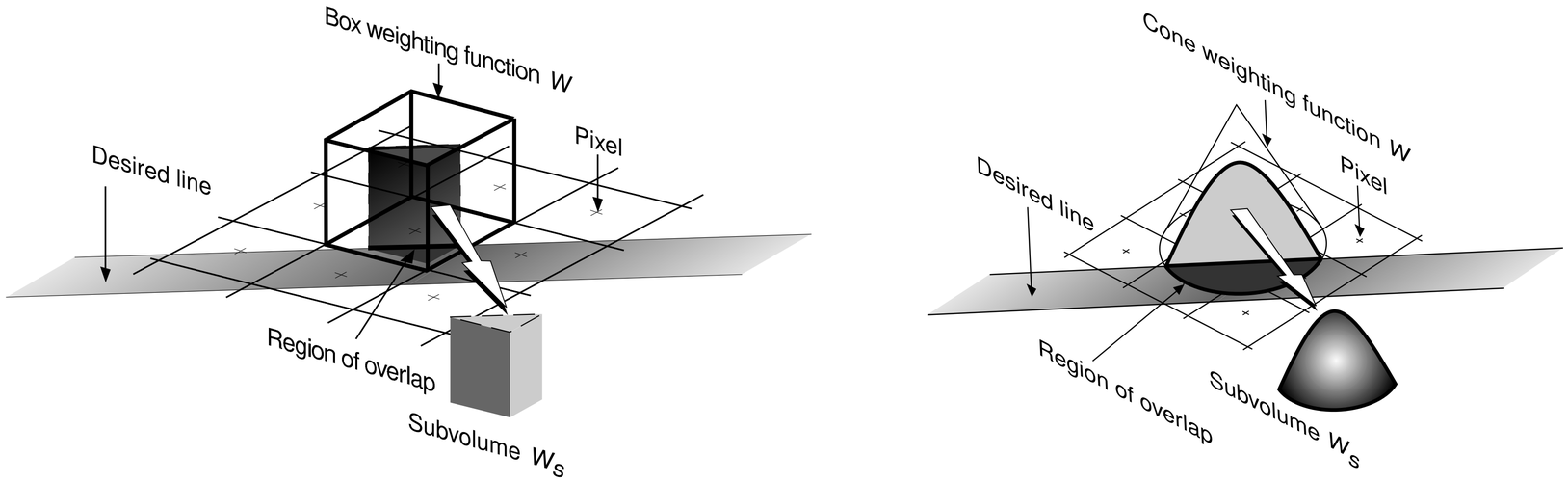
Line Width and Area

- Line has arbitrary width and occupies finite area on screen
  - discrete B/W pixels cannot approximate line area well
- Amount of distributed color (ink) per pixel can be adjusted to match intensity and area of line
  - using varying per-pixel color intensity depending on how much is covered by the line
  - smaller than pixel-width lines possible also possible



© [AW94] Figure 3.35,36

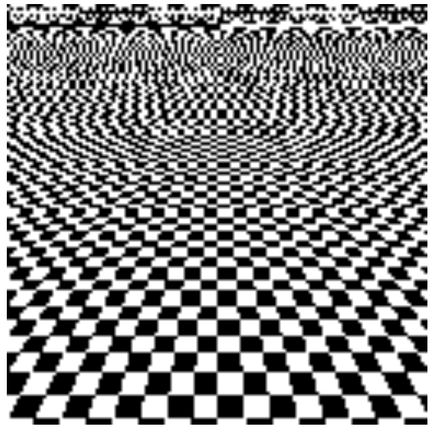
Area Sampling

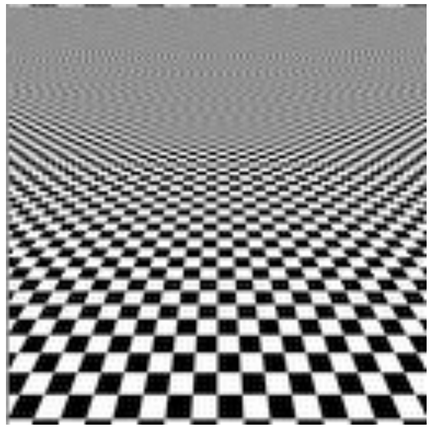


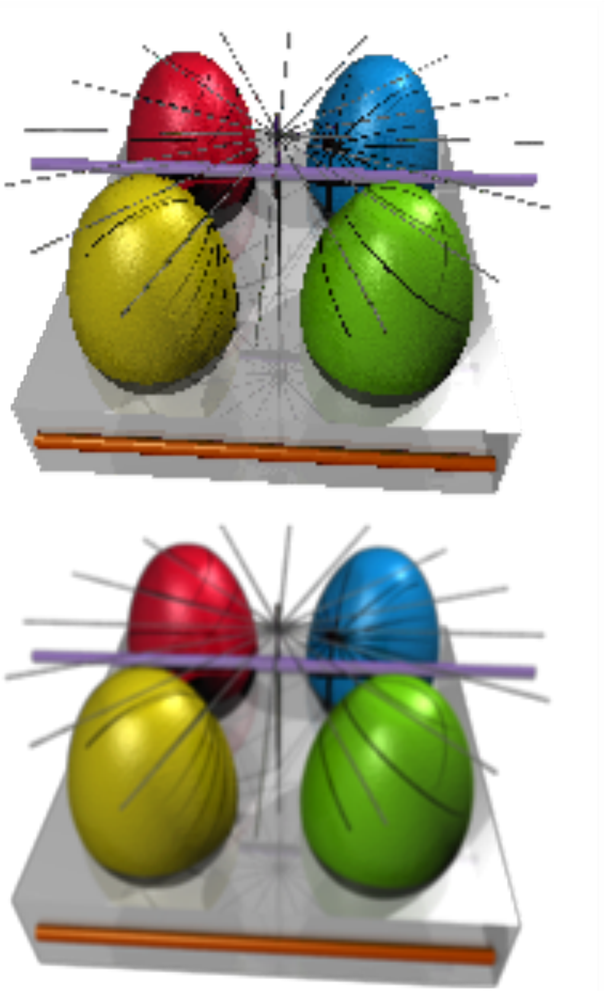
© [AW94] Figure 3.38,39

- Compute the partial area of a pixel covered by the line and color it proportionally to that
  - **unweighted**: intensity is equal to the % of the pixel's intersection area with the line
    - same intensity for equal area overlap
  - **weighted**: intensity is equal to the integral over the weighting kernel's intersection region with the line
    - overlap area closer to pixel center causes higher intensity

Area Antialiasing







- Texture antialiasing filters
  - multiple checkerboard squares contribute to the topmost pixels
- FSAA attempts to average the contribution of multiple color contributions per pixel
  - supersampling usually means that each frame is rendered at double (2x) or quadruple (4x) resolution
  - finally down-sampled to match the display resolution, 2x = four supersampled pixels

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